

Visualization Mapping Analysis of Emergency Management Scientific Research Big Data during COVID-19

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Abstract— [Objective] To better understand the research status of emergency management during the outbreak of COVID-19 in China, and further improve the level of China's emergency management, [method] In this paper, the big data of emergency management research during COVID-19 was visualized and analyzed, and the visual knowledge map was drawn to study the big data of China's emergency management literature. [Conclusion] This paper reveals the current situation and characteristics of China's emergency management during COVID-19 by using the visual knowledge map, summarizes the research trend in the future, and points out the new ways to improve China's emergency management ability in the period after the COVID-19.

Keywords-component; COVID-19; Big data of Scientific research; Emergency Management; CiteSpace

I. INTRODUCTION

Emergency management has become a hot topic of academic discussion and research during the period of the prevention and control of COVID-19. The application of big data in plague prevention and control has imperceptibly become the core force for emergency management. For better understand the research status of emergency management during the period of COVID-19 in China, by using bibliometric method and CiteSpace V software, this paper makes a statistical analysis of the literatures related to the emergency response capability during the COVID-19 period in China, draws a visual knowledge map, reveals the authoritative authors, the distribution of authoritative research institutions, the knowledge map of research the hotspots and cluster analysis, also makes constructive conclusions and suggestions on the future research trend.

II. DATA AND METHODS

A. Data sources

Academic research on emergency management during COVID-19 has been conducted on CNKI, Viper Information Network, CSSCI and China Academic Journal, etc. This paper studies the research trends related to emergency management during COVID-19 in China. Therefore, the author chooses the knowledge Network, which is influential and has been included for a long time and widely used by most domestic scholars, as the research field and data selection source.

B. Data collection

After entering CNKI on October 26, 2020, the author selected eight sub databases including journals, educational journals, characteristic journals, doctors, masters, domestic conferences, international conferences, newspapers, and academic journals. In the search conditions, the author entered the topic = new plague situation and theme = emergency management. 660 results were obtained. The data were cleaned and the nonacademic literatures such as comments were removed 651 final data results, data conversion, drawing knowledge map.

C. Research methods and software

For the past few years, literature big data has started to flood into the research fields of various disciplines. Among them, scient metrics and information metrology technologies have developed rapidly, providing a reliable way for big data visualization research. Knowledge graph is a visual analysis method of literature big data emerging in recent years. It has advantages such as large amount of literature processing, intuitive visualization, diverse analysis angles, high reliability of data analysis results and strong operability. Through the processing and visual analysis of literature big data, it has a unique contribution to the sorting out of the development context of a certain discipline, summarizing the research hotspots in a certain field and predicting the future research direction.

III. ANALYZE THE PROCESS AND RESULTS

A. The distribution and cooperative characteristics of Chinese research institutions

In the interface of CiteSpace 5.1[2], a visual analysis chart of institutional cooperation was performed on the relevant papers. The specific operations were: Time Slicing was 1, Node Types were selected as Institution, Links was selected as default, and top N was selected as threshold, where N=50. [3] Conducted quantitative analysis on the authors of relevant papers through CiteSpace and sorted out 86 nodes and 32 lines with a density of 0.0099.

The distribution and cooperation of scientific research institutions are obtained after clicking the operation. The nodes and fonts were adjusted appropriately, and the visual

atlas of the distribution and cooperation of scientific research institutions was obtained (Figure 1) :



Figure 1 collinear map of distribution and cooperation of scientific research institutions

number	organization	number	organization
1	Scientific Research Institute of the Ministry of transport	12	Department of Communication and Policy of Science Technology,USTC
2	School of society and politics, Anhui University	13	Wuhan Water Group Co.LTD
3	Nanchang Municipal Government	14	Central South China Municipal Engineering Design and Research Institute Co., Ltd
4	School of public administration, Tsinghua University	15	Zhangzhou People
5	Urban Transportation Planning Academic Committee of Chinese Urban Planning Society	16	China Science and Technology Research Institute
6	Nanjing City and Traffic Planning and Research Institute Co.LTD	17	China Institute of public welfare, Beijing Normal University
7	Tongji Hospital affiliated to Tongji Medical College of Science and Technology	18	General Hospital of Tianjin Medical
8	The Third Affiliated Hospital of Sun Yat-sen University	19	The First Affiliated Hospital of Anhui Medical
9	West China Hospital of Sichuan University	20	Disaster Prevention Committee of Chinese Urban Planning Society
10	Nanfang Hospital,Southern Medical University	21	Beijing Association for Disaster
11	School of Information Resource Management, Renmin University of China	22	China Disaster Prevention Association

In figure 1, the size of the organization name (nodes) reflects the agency published a new champions league during the outbreak of the number of emergency ability related papers, attachment between various agencies say the degree of mutual cooperation, through the above, the first is obvious: (1) institute of the Ministry of Transport (9 articles), department of science and technology communication and science and technology policy, University of science and technology of China (5 articles), and School of society and politics, Anhui University (5 articles) the number of the most; (2) most of them are departments or national scientific research units of some famous universities in China. Among them, West China Hospital of Sichuan University, Tongji Hospital of Huazhong University of Science and Technology, Tongji Hospital of Original dress, And Nanfang Hospital of Southern Medical University cooperated closely with the Third Affiliated Hospital of Sun Yat-sen University; the Department of technology communication and science and technology policy of University of science and technology of China cooperates closely with Anhui University, School of information resources management, Renmin University of China; Nanjing

City and Traffic Planning and Research Institute Co., LTD., the Urban traffic Planning Academic Committee of The Urban Planning Society of China and the National Academy of Development and Strategy of The People's Republic of China have close cooperation. As a result, three sub-networks are formed in many institutions, namely, medical institutions, transportation research institutions and intelligence and information research institutions. The author sorted out the top ranked institutions by frequency and centrality, as shown in Table 1:

Table 1 Research on the centrality of research institution distribution and cooperation

frequency	Center degrees	Author	Year
2 ⁺	5 ⁺	The First Affiliated Hospital of Anhui Medical University ⁺	2020 ⁺
2 ⁺	5 ⁺	General Hospital of Tianjin Medical University ⁺	2020 ⁺
2 ⁺	3 ⁺	Department of Library Information and Archives Management, School of Economics and Management, Chinese Academy of Sciences ⁺	2020 ⁺
2 ⁺	3 ⁺	Department of Information Resource Management, Business School, Nankai University ⁺	2020 ⁺
2 ⁺	3 ⁺	Documentation and information center of the Chinese academy of sciences ⁺	2020 ⁺
2 ⁺	3 ⁺	School of Public Administration, Renmin University of China ⁺	2020 ⁺
4 ⁺	2 ⁺	Tongji Hospital affiliated to Tongji Medical College of Huazhong University of Science and Technology ⁺	2020 ⁺
3 ⁺	2 ⁺	West China Hospital of Sichuan University ⁺	2020 ⁺
3 ⁺	2 ⁺	Nanjing City and Traffic Planning and Research Institute Co. LTD ⁺	2020 ⁺
3 ⁺	2 ⁺	Nanfang Hospital, Southern Medical University ⁺	2020 ⁺
5 ⁺	1 ⁺	School of Social and Political Science, Anhui University ⁺	2020 ⁺
4 ⁺	1 ⁺	China Municipal Engineering Central South Design and Research Institute Co. LTD ⁺	2020 ⁺
9 ⁺	0 ⁺	Institute of Science, Ministry of Transport ⁺	2019 ⁺
5 ⁺	0 ⁺	Department of Communication and Policy of Science and Technology, USTC ⁺	2020 ⁺
4 ⁺	0 ⁺	School of Public Administration, Tsinghua University ⁺	2020 ⁺
4 ⁺	0 ⁺	China Science Research Institute ⁺	2020 ⁺

The greater the centrality, the more important it is in the network. The chart shows that (1) general hospital of Tianjin medical university, and the first affiliated hospital of Anhui medical university center degrees are 5, secondly, Chinese Academy of Sciences, college of economics and management books information and archives management, business school of nankai university information resources management, document information center, Chinese Academy of Sciences center associated with the school of public administration, Renmin university of China degree is higher, also visible during the outbreak of the new champions league in general hospital of Tianjin medical university and the first affiliated hospital of Anhui medical university, led by the medical resources is given priority to, with the Chinese Academy of Sciences at the university of economics and management college library and archives management information intelligence unit is complementary,It has formed a core scientific research institution during the COVID-19 plague;(2) The academy of Sciences of the Ministry of Transport has published the most papers, most of which focus on the emergency transport system, comprehensive law enforcement capacity of transport, construction of transport laws and

B. Lead author and co-research trends

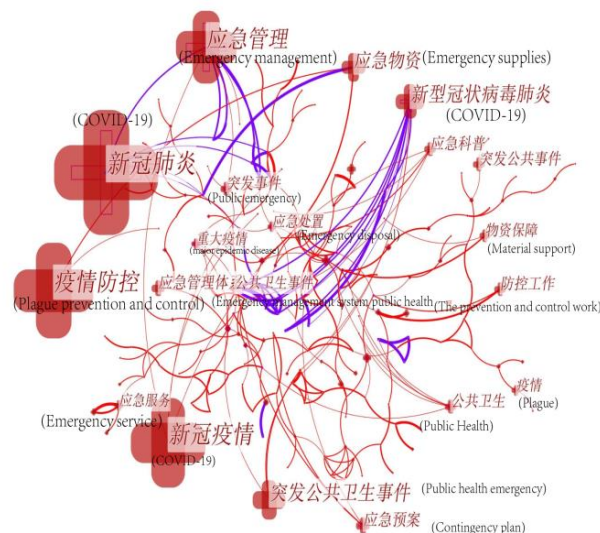
OneSpace, v. 6.0.8.0 (22Apr)
September 15, 2020 3:12:12 PM CST
Web: F-Index.org/2020/9/15/oneSpace.html
Timestamp: 2075-2021 (Silence Length=1)
Selection Criteria: gindex (w=0.5), LRF=0.5, LRF=0.5, LRF=0.5, w=0.0
Network: 1000, 1000 (w=0.5, density=0.0126)
Largest CC: 100 (22.0%)
Nodes Labeled: 2.0%
Pruning: Pathfinder

张叔林 (Zhang Shulin)
卢文刚 (Lu Wengang)
习近平 (Xi Jinping)
庞彪 (Pang Biao)
沈波 (Shen Bo)
张芳 (Zhang Fang)
刘畅 (Liu Chang)
龚箭 (Gong Jian)
尚政民 (Shang Zhengmin)
徐开元 (Xu Kaiyuan)
张建斌 (Zhang Jianbin)
顾均 (Gu Jun)
杜志杰 (Du Zhizhe)
李帅 (Li Shuai)
李德华 (Li Dehua)
刘美玲 (Liu Meiling)
李宝华 (Li Baohua)
徐双能 (Xu Shuangneng)
吴双能 (Wu Shuangneng)
徐由海 (Xu Youhai)
张强 (Zhang Qiang)
任晓蕊 (Ren Xiaorui)
戴磊 (Dai Lei)
丁玉祥 (Ding Yuxiang)
李燕霞 (Li Yanxia)
欧阳斌 (Ouyang Bin)
周艾燕 (Zhou Aiyen)
丁芝华 (Ding Zhihua)
安国俊 (An Guojun)
朱建波 (Zhu Jianbo)
贾超玮 (Jia Chaowei)
杨涛 (Yang Tao)
李佳柔 (Li Jiarou)

Lead author and co-author of a visual atlas of research trends In Figure 2, the size of each scholar's name represents the frequency of their cooperation, and the link between the authors indicates the cooperation between them. Among the authors of the paper on the emergency response capability of the NOVEL Coronavirus plague in the above figure, General Secretary Xi Jinping ranks the first, reflecting the core strength of national leaders during the plague response period. In addition, Liu Chang (7 articles), Gong Jian (6 articles) and Yang Chen (5 articles) are among the top producers in terms of output, and they work closely together. Among them, Xia Ouyang, He Jing, Wu Wei, Gu Xue Kui, and Sun Yueling 5 authors have cross-cooperation and close communication. The authors ranked the top authors according to their publication volume, as shown in Table 2:

Quantitative ¹	Author ²	Year ³	Quantitative ¹	Author ²	Year ³
9 ³	Xi Jinping ³	2020 ³	3 ³	Shen Bo ³	2020 ³
7 ³	Liu Chang ³	2019 ³	3 ³	Yang Tao ³	2020 ³
6 ³	Gong Jian ³	2020 ³	3 ³	Du Zhijie ³	2020 ³
5 ³	Yang Chen ³	2019 ³	3 ³	Li Yanxiao ³	2020 ³
4 ³	Ou Ynagbin ³	2020 ³	3 ³	Li Dehua ³	2020 ³
3 ³	Gu Yun ³	2020 ³	3 ³	Xu Kaiyuan ³	2020 ³
3 ³	Jia Fuwei ³	2020 ³	3 ³	Zhang Jianbin ³	2020 ³

Keywords co-occurrence knowledge map can be used to reveal the characteristics more comprehensively and systematically, key research areas, knowledge transfer, research frontiers and research trends of emergency management during COVID-19. A Node Type is selected as Keyword, the pruning method is path, and other parameter Settings remain unchanged. Click the run program and enter the Visualization interface [4]. In this paper, the key words of 651 articles are counted, followed by integration based on the similarity of keywords.



As can be seen from Figure 3, the number of nodes in the key word map under the parameter setting condition N=206, indicating that 206 high-frequency co-occurrence keywords are included in the statistical results. Line number E=272; Network Density (Density) =0.0129; The node size represents the occurrence frequency of keywords, and the line thickness represents the co-occurrence intensity. In order to present the new crown outbreak during emergency management research, the author statistics the keywords co-occurrence high

intermediary centrality keywords of knowledge map, as shown in figure 3, we can see about the new champions league during plague emergency management research around the new outbreak and emergency management two aspects, including emergency management including emergency supplies, emergency popular science, emergency disposal and emergency services, new crown plague research including public health emergencies, prevention and control work, COVID - 19, etc. From the perspective of high intermediary centrality distribution, there are health emergency management, closed management, medical treatment, and temperature measurement, etc. Therefore, the research focus of emergency management during COVID-19 is on the issues related to medical prevention and control measures. The key words in the figure above, whose occurrence frequency and centrality rank first, are sorted out as shown in Table 3

Table 3 Statistical table of keywords with high mediating centrality

Frequency	Center degrees	Betweenness centrality	Keyword	Frequency	Center degrees	Betweenness centrality	Keyword
8 ⁽¹⁾	8 ⁽¹⁾	0.4 ⁽¹⁾	Health emergency management ⁽¹⁾	3 ⁽¹⁾	3 ⁽¹⁾	0.22 ⁽¹⁾	Countermeasures and Strategies ⁽¹⁾
5 ⁽¹⁾	5 ⁽¹⁾	0.37 ⁽¹⁾	Closed management ⁽¹⁾	4 ⁽¹⁾	3 ⁽¹⁾	0.21 ⁽¹⁾	Traffic control ⁽¹⁾
8 ⁽¹⁾	5 ⁽¹⁾	0.34 ⁽¹⁾	Medical treatment ⁽¹⁾	17 ⁽¹⁾	6 ⁽¹⁾	0.19 ⁽¹⁾	Public health ⁽¹⁾
2 ⁽¹⁾	4 ⁽¹⁾	0.31 ⁽¹⁾	Temperature measurement ⁽¹⁾	2 ⁽¹⁾	5 ⁽¹⁾	0.18 ⁽¹⁾	Emergency supplies rescue ⁽¹⁾
14 ⁽¹⁾	5 ⁽¹⁾	0.27 ⁽¹⁾	Major outbreaks ⁽¹⁾	92 ⁽¹⁾	8 ⁽¹⁾	0.17 ⁽¹⁾	Emergency management ⁽¹⁾
13 ⁽¹⁾	5 ⁽¹⁾	0.27 ⁽¹⁾	Emergency response ⁽¹⁾	15 ⁽¹⁾	7 ⁽¹⁾	0.17 ⁽¹⁾	Public health event ⁽¹⁾
2 ⁽¹⁾	8 ⁽¹⁾	0.26 ⁽¹⁾	Clinical Waste management ⁽¹⁾	5 ⁽¹⁾	2 ⁽¹⁾	0.17 ⁽¹⁾	The prevention and control strategy ⁽¹⁾
2 ⁽¹⁾	5 ⁽¹⁾	0.26 ⁽¹⁾	Close contact ⁽¹⁾	6 ⁽¹⁾	3 ⁽¹⁾	0.16 ⁽¹⁾	Colleges and universities ⁽¹⁾
2 ⁽¹⁾	7 ⁽¹⁾	0.25 ⁽¹⁾	Public transport ⁽¹⁾	2 ⁽¹⁾	4 ⁽¹⁾	0.16 ⁽¹⁾	Emergency information ⁽¹⁾
2 ⁽¹⁾	5 ⁽¹⁾	0.25 ⁽¹⁾	Assistance system ⁽¹⁾	45 ⁽¹⁾	10 ⁽¹⁾	0.15 ⁽¹⁾	COVID-19 ⁽¹⁾
9 ⁽¹⁾	6 ⁽¹⁾	0.24 ⁽¹⁾	Big data ⁽¹⁾	9 ⁽¹⁾	4 ⁽¹⁾	0.15 ⁽¹⁾	Community workers ⁽¹⁾
6 ⁽¹⁾	4 ⁽¹⁾	0.24 ⁽¹⁾	Xi Jinping ⁽¹⁾	17 ⁽¹⁾	5 ⁽¹⁾	0.14 ⁽¹⁾	The emergency ⁽¹⁾
3 ⁽¹⁾	4 ⁽¹⁾	0.23 ⁽¹⁾	Medical observation ⁽¹⁾	14 ⁽¹⁾	5 ⁽¹⁾	0.14 ⁽¹⁾	The emergency ⁽¹⁾

D. Visualization of the development and change characteristics of research hotspots

Clustering refers to the process of dividing a collection of abstract objects into several classes composed of similar objects². A cluster generated by a cluster is a collection of data objects. CiteSpace software provides automatic clustering⁴. After automatic clustering, CiteSpace extracted clustering label words based on three sorting algorithms. The clustering label words were derived from cited literature. In this paper, log-likelihood rate was selected to extract clustering label, which emphasizes more on the characteristics of the research field. On the basis of the cooccurrence knowledge map of key words in emergency management during COVID-19, the clustering function of CiteSpace software was used to extract and display cluster labels at the same time, and the clustering knowledge map of key words in emergency management during COVID-19 was obtained as shown in Figure 4. The figure 4 shows that CiteSpace map of clustering analysis, Q value (module) is 0.5401, S value (contour value) is 0.6458, there're a total of eight map clustering, respectively is emergency management, # 1 # 0 plague prevention and control, emergency disposal # 2, # 3, # 4 knowledge services, public health # 5, # 6 COVID-19 - 19 in colleges and universities, emergency management system and # 7 # 8 archives management. Further statistical analysis is carried out for each cluster, and the results are shown in Table 4:

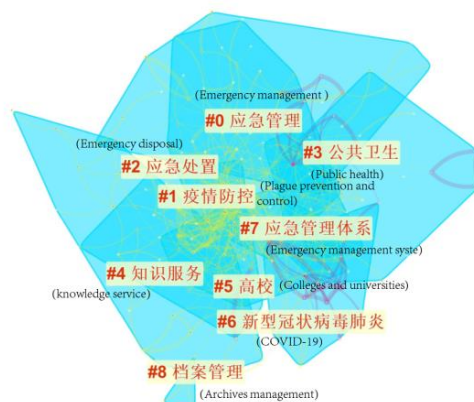


Figure 4. Knowledge map of key words clustering in emergency management research during COVID-19

Table 4 Cluster list of key words in emergency management research during COVID-19

Clustering ID	Number of cluster members	Cluster tags	Cluster contour values	LLR ⁽¹⁾
0 ⁽¹⁾	35 ⁽¹⁾	0.819 ⁽¹⁾	Emergency management	Emergency management; Emergency services; Popularization of emergency science; Emergency supplies; Emergency broadcasting; Emergency plan; Medical treatment; New media; Emergency language services; Sina; Weibo; Emergency response; Emergency management of public opinions; Management of emergency supplies; Emergency command system; Emergency diplomacy; Intelligence resource allocation; Emergency measures; Big data capability construction; Think tank emergency; Emergency information ⁽¹⁾
1 ⁽¹⁾	34 ⁽¹⁾	0.875 ⁽¹⁾	Plague prevention and control	Plague prevention and control; COVID-19; Community workers; Health emergency management; Medical treatment; Closed management; Designated hospitals; Medical insurance for serious diseases; Salvage system; General clinic; Big data; Psychological assistance hotline; Medical treatment process; Medical equipment; Prevention and treatment by the masses; Large grade A hospitals; Green channel; Primary medical and health institutions; County-level CDC ⁽¹⁾
2 ⁽¹⁾	31 ⁽¹⁾	0.817 ⁽¹⁾	Emergency disposal	Emergency response; Traffic control; Medical observation; Isolation observation; Key points of control; Disposal plan for wearing masks; First diagnosis responsibility system; Waste classification management; Nucleic acid testing; Isolation measures; Medical assistance; Liquid soap; Key links; Government directives; Key areas; Close contacts; Specialized hospital for infectious diseases; Big data; Disinfection treatment; Persons with the virus; Community worker ⁽¹⁾
3 ⁽¹⁾	27 ⁽¹⁾	0.761 ⁽¹⁾	Public health	Public health; disinfection standard; intelligence; plague monitoring; administrative compulsion; social security; overseas public health strategic intelligence; blood collection and supply; new coronavirus; public health safety; big data; community workers; public health events; medical personnel; infectious disease prevention and control law; public health services; medical observation; medical waste management; joint prevention and control; Internet rumors ⁽¹⁾
4 ⁽¹⁾	23 ⁽¹⁾	0.832 ⁽¹⁾	Knowledge service	Knowledge service; emergency novel coronavirus pneumonia prevention and control; supply chain; emergency logistics system; grassroots emergency mobilization; big data technology; crisis communication medical guidance; protective gloves; crisis management; emergency science popularization; medical treatment; big data; community workers; public health service; medical personnel; infectious disease prevention and control; infectious disease hospital; medical observation ⁽¹⁾
5 ⁽¹⁾	21 ⁽¹⁾	0.857 ⁽¹⁾	Colleges and universities	Universities; emergency management mechanism; polluted area; medical service process; ideological and political education; emergency medical; clean area; hospital logistics management; fire rescue; prevention and control plan; emergency thinking; emergency science popularization; public health; medical treatment; big data; emergency disposal; closed management; traffic control; material support; medical observation; body temperature measurement; Internet rumors ⁽¹⁾
6 ⁽¹⁾	15 ⁽¹⁾	0.909 ⁽¹⁾	COVID-19	Novel coronavirus novel coronavirus pneumonia; emergency medical supplies scheduling; public crisis warning; plague prevention and control; blood collection and supply organization; disease control and emergency work; plague prevention materials; intelligence capability; big data governance; blood donors crowd characteristics; new crown pneumonia material barrier; emergency science popularization; community workers; health emergency management; closed management; infectious disease hospital; medical observation; body temperature measurement ⁽¹⁾

7 ^{c1}	9 ^{c1}	0.825 ^{c1}	Emergency management system ^{c1}	Emergency management system; risk society; panic; new national system; security system; innovation and development; regionalization; risk diffusion; plague prevention and control; informatization; emergency management; material support; emergency service; emergency science popularization; big data; emergency response mechanism; public library; traffic control command system; emergency system; emergency material reserve; emergency logistics; ^{c1}
8 ^{c1}	3 ^{c1}	0.993 ^{c1}	File management	Archives management; public crisis management; plague prevention and control; emergency science popularization; medical treatment; big data; medical personnel; public crisis; emergency management; infectious disease hospital; traffic control emergency system; medical observation; logistics management; medical waste management; joint prevention and control; public hospital; emergency logistics; emergency industry; architectural design; legal guarantee; urban governance; ^{c1}

Table 4 lists the ID of each cluster, the number of cluster members, the contour value of the cluster, the average year, and the cluster words extracted based on LLR algorithm that are highly correlated with the cluster. It can be concluded that the cluster of #0 emergency management mainly refers to a series of actions taken from traffic control and material storage after COVID-19 outbreak. #2 Cluster of emergency response mainly refers to specific prevention and control measures and measures taken during plague prevention and control; The cluster of #7 emergency management system mainly refers to the construction of emergency management system from the overall perspective in the whole prevention and control process, including the summary of problems and the proposal of strategies.

In addition, it can be concluded that: #1 Cluster of plague prevention and control mainly refers to people involved in plague prevention and control from the perspective of medicine, prevention, and control hospitals, etc. #6 COVID-19 reflects research related to COVID-19 from media, medical, prevention and control and psychological perspectives.

We can also see that the cluster contour value of the #8 file management cluster is the highest, indicating that its clustering results are better. In the whole process of COVID-19 prevention and control, archives management has become an important management tool throughout the whole process, from emergency management, logistics material allocation, to medical treatment, traffic control, public crisis response, to knowledge service dissemination and guidance. 5. Specific behaviors of the special group of colleges and universities reflected by the cluster of colleges and universities in plague prevention and control; The contour value of knowledge service cluster is not high, but during COVID-19, the output and contribution of knowledge service platform are needed for key measures, including grassroots emergency mobilization, emergency popularization, closed management, intelligent travel and isolated observation, etc., so knowledge service plays a role of good medicine during the plague prevention and control.

IV. DISCUSSION

From the above analysis of emergency management related papers during the new crown outbreak with CiteSpace, the following can be seen.

First, from the distribution and cooperation of scientific research institutions in China, we can see that numerous research institutions have formed three sub networks, which are respectively medical institutions, traffic research

and intelligence information research institutions, and have become a frontier for emergency management research during outbreak control. Based on medical resources such as Tianjin

Medical University General Hospital and the First Affiliated Hospital of Anhui Medical University, supplemented by information intelligence units such as Department of book intelligence and archive management, school of economics and management, Chinese Academy of Sciences, this form the core scientific research institution during the new crown outbreak. Second, from the primary author and the co research landscape, it is obvious to see Xi's general secretary ranked first, significantly more than the other authors, reflecting the central power of national leaders during the plague.

In addition, Liu Chang (7 articles), Gong Jian (6 articles) and Yang Chen (5 articles) are among the top producers in terms of output, and they cooperate closely. Among them, Xia Ouyang, He Jing, Wu Wei, Gu Xuekui, he Sun Yueling 5 authors have cross-cooperation. Again, from the point of view of research hot spots, it is organized around the new crown outbreak, emergency management core keywords, of which the research heat on three key words of "closed management", "medical care", and "body temperature measurement" is high. It indicated that medical emergency disposal measures maintained sustained enforcement ability during the outbreak control period, which reflects a strong emergency disposal ability during the outbreak control period in China.

Finally, according to the prevention and control process of outbreaks and CiteSpace clustering diagram, the high clustering profile values in colleges indicate that the prevention and control work in colleges has been well conducted; archive management has been continuously concerned during the emergency management of outbreaks, and insufficient file management directly affects the emergency services insufficient capacity, and the information files have not kept up in time and update, there has been no timely and effective update to historical materials and information, and emergency services capabilities have been impacted to some extent. Such as the incomplete collection of data, charts, mapping and so on for various types of major accidents and natural disasters in our country's calendar year, the classification and sorting is not standardized, which is not conducive to the support of data in emergency work, and the advance of knowledge services into the wading field of emergency management in outbreaks has a crucial impact on promoting the effectiveness and popularity of emergency management. As it spread during the plague, there was a high demand for authoritative and comprehensive knowledge about outbreak control among the national population. It is mainly focused on the following aspects: 1. Outbreak control policy (national policy publication, etc.); 2. Outbreak control real-time information (medical and medical rescue, etc.); 3. Outbreak protective medical information (popular science knowledge such as masks, disinfection, etc.); information (suspension without suspension, etc.). Meanwhile, "big data technology" appeared in eight clusters, which indicated that big data has penetrated invisible to the process of emergency management during an outbreak, big data provides data support for emergency management decisions, emergency management systems and emergency disposal, is a data support for the study of outbreak control and new coronal pneumonia, and at the same time is a practical tool for archive management, an important channel of knowledge services, So big data is

very important for enhancing emergency management research during the new crown outbreak.

V. Conclusion

Current literature is mainly through the way of literature research to explore the aspects of information construction, emergency disposal wording, intelligence pathways and big data applications of emergency management during an outbreak, but there has been no quantitative research on this topic through the way of statistical analysis.

In this paper, after affirming the huge contribution and among the inadequacies of the existing literatures, the CiteSpace software was used to extract the relevant literatures of Chinese knowledge, and deeply mine their relevant scientific research institutions, main authors, keywords, and other elements, to derive and solve the research on how emergency management was carried out during the new crown plague. After a discussion and induction of the findings, in this paper, aiming at the characteristics of studies related to emergency management during outbreak control and prevention, we put forward the following concluding suggestions:

First, big data technology is a major path to boost emergency management during an outbreak. Many research institutes for emergency management during the outbreak control period focus on research departments such as medical, transportation and intelligence, which is in accordance with the dynamic division of key areas during the outbreak control period. Through cluster mapping analysis, it is known that big data technology is used in the whole process of outbreak control and prevention, so the author suggests using the field of computer information as the starting point, big data technology to improve emergency management research during the outbreak will certainly become a future research hotspot and development trend, at the same time, should be introduced into multi-domain, multi-disciplinary cross-over research to better improve the emergency management system and improve. Second, archives management is a new force in the study of emergency management during the plague, which is the key point of the future emergency management system construction. With information technology being widely used in all walks of life, electronic archives replace the previous paper materials, break through many key technologies such as data integration and resolution of diagnostic strategies that cannot be achieved by paper materials, and basically realize the interoperability and intelligence of data[5]. Based on the traditional archive management within the emergency management system, to explore the informatics work of archive management in emergency management under the new situation will certainly be a key point in the construction of emergency management systems for unexpected events in the future.

Finally, knowledge services became the highlight of emergency management research during the outbreak. Based on the need for information for prevention and control of the new corona pneumonia plague, To analyze the strategies of emergency popular science and health prevention and control in the context of the new corona pneumonia outbreak control, to construct a complete emergency knowledge resource system for outbreak control and prevention, to build up a platform for

intelligent knowledge service of emergency index for outbreak control, to improve the top management mechanism of emergency knowledge service for outbreak control and prevention, and to establish a long-lasting service mechanism for emergency management of outbreak control and prevention, with the aim of identifying possible public health emergencies in the future Knowledge service provision needed for emergency management is referenced and borrowed^{[6][7][8]}.

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Reference

- [1] Chaomei Chen (author), Yue Chen (transl), Jianhua Hou (transl), Yongxia Liang (transl). CiteSpace II: identification and visualization of new trends and neodynamics in the scientific literature [J]. *Journal of intelligence*, 2009 (3): 401-421
- [2] Liu D. analysis of the developmental course of subject domain research based on CiteSpace atlas using Chinese education in Thailand as an example [J]. *Modern vocational education*, 2020 (26): 32-34
- [3] Jiang B, Qing Liu D. analysis of the current status of domestic scholars on the study of Chinese education in Malaysia - CiteSpace based visual analysis [J]. *Journal of Kunming academy*, 2019,41 (04): 106-114
- [4] Liu Di, Zhang Yanqi, Yin Tao. A CiteSpace based analysis of research hotspots and subject evolution in high plateau lakes in China [J]. *Yunnan hydraulic power generation*, 2019,35 (03): 22-25
- [5] Ling Zhang. Analysis of the construction of archive management informatics for emergency management systems [J]. *Safety production in China*, 2020, 15 (03): 40-41
- [6] Junsong Bai. Analysis of emergency management of emergency response dossiers [J]. *Ingenuity*, 2020 (08): 231
- [7] Funding project: the "" study on the emergency capability of public safety improvement by big data technology in new crown outbreak control and prevention "" (Project No. 2017zzx251), a scientific research foundation project of Yunnan Provincial Department of Education
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